

Engineering Department Doc, Layer 2 v0.2.3 Shape Metric Quality Flags and Layer 3 Readiness Filter

Final Layer 2 cleanup update after team_width/team_depth patch and quality-gate QA

Purpose: This document records the final Layer 2 v0.2.3 cleanup result. It adds a non-destructive quality flag table for current Layer 2 shape metrics so Layer 3 can consume shape features with explicit warning and inclusion logic. It does not authorize Layer 3 modeling, tactical identity profiles, AI interpretation, dashboards, reports, recommendations, or advanced tactical claims.

Current decision: Layer 2 is now ready to hand off to Layer 3 v0.1 descriptive tactical profiling, provided the next developer consumes the shape quality flags and reports quality counts. Layer 3 should not silently ignore warnings or treat flagged rows as if they were unqualified evidence.

1. Context from v0.2.1 and v0.2.2

- Layer 2 v0.2.1 reviewed the original team_width warning and found a team_width/team_depth semantic mismatch.
- Layer 2 v0.2.2 patched the metric definition so team_width uses the lateral y_metric range and team_depth uses the longitudinal x_metric range.
- After the v0.2.2 patch, Layer 2 QA passed and football sanity QA passed.
- The remaining geometry issue was small and residual: 256 strict team_width pitch-bound warnings out of 873,296 frame-team rows; there were 0 width tolerance failures.
- Layer 2 v0.2.3 converts that residual issue into a non-destructive quality flag workflow for Layer 3 readiness.

2. Files Created or Changed

File	Purpose
src/soccer_tips/features/shape_quality.py	Builds the shape metric quality flags table from frame_team_shape_metrics.csv. Adds geometry flags, player/data quality flags, warning counts, exclusion reasons, and layer3_shape_metric_include.
src/soccer_tips/features/qa_shape_quality.py	Command-line QA runner for the quality flags table. Writes layer2_shape_metric_quality_flags.csv and layer2_shape_metric_quality_summary.json.
tests/test_shape_quality.py	Lightweight tests for flag calculation, warning-vs-exclusion behavior, and Layer 3 include logic.

3. Output Files Created

Output	Description	Row grain
data/processed/features/layer2_shape_metric_quality_flags.csv	Non-destructive quality flags for current Layer 2 frame-team shape metrics. Joinable back to frame_team_shape_metrics.csv.	One row per provider, match_id, frame, period, and team_id.
data/processed/features/layer2_shape_metric_quality_summary.json	Structured QA summary for the quality flags table and Layer 3 include/exclude counts.	One JSON object per QA run.

4. Join Keys and Context Fields

- Required join keys: provider, match_id, frame, period, team_id.
- Included context fields: phase_index, possession_status_for_team, team_in_possession_phase_type, team_out_of_possession_phase_type.
- Included reference fields: pitch_length, pitch_width, team_width, team_depth, player_count_used, detected_player_count, extrapolated_player_count.

5. Flag Definitions and Threshold Rules

Flag	Rule	Layer 3 treatment
team_width_strict_bound_flag	team_width > pitch_width	Warning only. Not an automatic exclusion.
team_width_tolerance_fail_flag	team_width > pitch_width * 1.05	Hard exclusion.
team_depth_strict_bound_flag	team_depth > pitch_length	Warning only. Not an automatic exclusion.
team_depth_tolerance_fail_flag	team_depth > pitch_length * 1.05	Hard exclusion.
low_player_count_flag	player_count_used < 8	Hard exclusion.
high_extrapolated_count_flag	extrapolated_player_count >= 6	Warning only. Not an automatic exclusion.
high_extrapolated_rate_flag	extrapolated_player_count / player_count_used >= 0.50	Warning only. Not an automatic exclusion.

6. layer3_shape_metric_include Rule

Initial rule: layer3_shape_metric_include is False only when team_width_tolerance_fail_flag is True, team_depth_tolerance_fail_flag is True, or low_player_count_flag is True.

- Strict width/depth bound flags remain caution flags and do not automatically exclude rows.
- High extrapolated count and high extrapolated rate remain caution flags and do not automatically exclude rows.
- This design keeps all Layer 2 metrics intact while giving Layer 3 a clear safety gate.

7. v0.2.3 Validation and QA Results

Check	Result
tests/test_layer2_features.py	All Layer 2 feature sanity checks passed.
tests/test_football_sanity.py	All Layer 2 football sanity tests passed.
tests/test_shape_quality.py	All Layer 2 shape quality tests passed.
Layer 2 SkillCorner batch build	Pass across all 10 processed SkillCorner matches.
Layer 2 QA	Status: pass. 873,296 frame-team rows, 9,104 phase-segment rows, 394 match-profile rows.
Football sanity QA	Status: pass. team_width_pitch_bound, team_depth_pitch_bound, convex_hull, negative values, player count, detection quality, and phase context all passed.
Shape metric quality QA	Status: pass.

Shape Quality Summary Field	Result
Rows checked	873,296
Rows included for Layer 3	873,296
Rows excluded from Layer 3	0
Exclusion percentage	0.000000%

Warning percentage	29.576913%
Strict width warnings	256
Width tolerance failures	0
Strict depth warnings	0
Depth tolerance failures	0
Low player count rows	0
High extrapolated count rows	258,068
High extrapolated rate rows	258,068

Status reason: No tolerance failures or low-player-count exclusions were found. Strict-bound and extrapolation flags remain available as non-destructive warnings.

8. Interpretation for Layer 3

- The team_width/team_depth semantic mismatch has been patched and revalidated.
- The residual strict width warnings are rare and do not exceed the 5% tolerance threshold.
- No rows are currently excluded from Layer 3 by the v0.2.3 include rule.
- The high warning percentage is driven by extrapolation flags, which reflect broadcast tracking quality caution, not automatic geometry failure.
- Layer 3 may begin descriptive tactical profiling, but it must join and consume layer2_shape_metric_quality_flags.csv before using shape metrics.
- Layer 3 outputs should report quality counts: included rows, excluded rows, strict-bound warnings, extrapolation warnings, and any tolerance failures.

9. Final Layer 3 Readiness Recommendation

- Ready to begin Layer 3 v0.1 - Descriptive Tactical Profiling.
- Start with conservative descriptive profiles: team shape tendencies by phase, match/team summaries, variation/stability summaries, and simple team comparisons.
- Do not begin clustering-first modeling, AI interpretation, dashboards, reports, recommendations, or final tactical identity claims yet.
- Do not use shape metrics without joining the quality flags table.
- Do not ignore extrapolation warnings; they should appear in Layer 3 quality metadata even when rows are included.

10. General Developer 1 Retirement Decision

Decision: General Developer 1 can now retire cleanly after Lead Developer acceptance of this v0.2.3 final update. Layer 2 has a non-destructive Layer 3 readiness filter, the semantic width/depth bug is patched, and the remaining residual quality concerns are documented as flags rather than hidden filtering.

11. Scope Guard

- No canonical Layer 2 schemas were changed.
- No Layer 2 rows were deleted or silently filtered.
- No Layer 3 outputs were created.
- No tactical identity profiles were created.
- No AI interpretation, dashboards, reports, recommendations, or advanced tactical claims were created.